MONEY MATH · DERIVATION

Where e comes from in compound interest

A companion to the one-page compound interest sheet

1 The question

A balance P earns an annual rate r, compounded n times a year. After t years it is

$$A_n(t) = P\left(1 + \frac{r}{n}\right)^{nt}. \tag{1}$$

Compounding more often always helps, but by less each time. Does $A_n(t)$ grow without bound as $n \to \infty$, or does it settle?

Definition 1. Euler's number is the limit $e = \lim_{m\to\infty}\left(1 + \frac{1}{m}\right)^{m} \approx 2.71828.$

2 The limit

Theorem 1. For every $r > 0$ and $t \geq 0$, $\lim_{n\to\infty} A_n(t) = Pe^{rt}$.

Proof. Substitute $m = n/r$, so $r/n = 1/m$ and $nt = m\,rt$. Then

$$A_n(t) = P\left(1 + \frac{1}{m}\right)^{m\,rt} = P\left[\left(1 + \frac{1}{m}\right)^{m}\right]^{rt}.$$

As $n \to \infty$, $m \to \infty$ too. The inner bracket tends to e, and $x \mapsto x^{rt}$ is continuous, so the whole expression tends to Pe^{rt}. □

3 How fast the gap closes

Taking logarithms and expanding $\ln(1+x) = x - \frac{x^2}{2} + O(x^3)$ with $x = r/n$:

$$\ln A_n(t) = \ln P + nt\left(\frac{r}{n} - \frac{r^2}{2n^2} + O(n^{-3})\right) = \ln P + rt - \frac{r^2 t}{2n} + O(n^{-2}). \tag{2}$$

So the shortfall from continuous compounding is roughly a factor $e^{-r^2 t/2n}$: halving it means doubling n.

Problem 1. At $r = 5\%$ for $t = 30$ years, how much more does continuous compounding pay than monthly ($n = 12$) on \$10,000?

Answer. $10{,}000\,e^{1.5} - 10{,}000\left(1 + \frac{0.05}{12}\right)^{360} \approx 44{,}816.89 - 44{,}677.44 \approx \$139$, in line with $e^{-r^2 t/2n} = e^{-0.003125} \approx 0.9969$ from the estimate above.